Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	A technology for growing and integrating III-V optoelectronic devices on arbitrary platforms			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum (√)			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics			
	Medium Category	Small Category	Medium Category	Small Category
	Semocinductor Devices and Systems	Compound Semiconductor Devices		
Concept of Proposed Technology	This proposal advances a III–V optoelectronic technology platform built on two core concepts: high-quality epitaxial growth of III–V device layers and their subsequent decoupling and heterogeneous integration onto diverse host platforms. First, high-quality, wafer-scale III–V epilayers are grown with precise control of composition, thickness, and interfaces. The III–As/P devices, which we can grow, cover infrared emitters, detectors, and photovoltaic structures with high optical and electronic performance. Second, these epilayers are separated from their growth substrates using sacrificial-layer-based exfoliation, including epitaxial lift-off, graphene-assisted remote epitaxy, and 2D-assisted lift-off. This decoupling step enables substrate reuse and allows the transfer and integration of III–V devices onto various platforms including silicon, glass, and flexible platforms. The resulting capability provides a modular pathway from epitaxial materials to system-level integration. Korean SMEs are expected to leverage these transferable III–V optoelectronic building blocks to develop application-specific systems such as chemical and biosensors, LiDAR and optical ranging modules, energy-harvesting devices, and emerging physical-AI hardware, accelerating translation from materials to deployable technologies.			

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Rationale for Joint R&D Project	Deployment of III–V optoelectronic systems require both device-grade epitaxial materials and manufacturing-compatible pathways for integration onto non-native platforms. While world-class capabilities exist in thin-film epitaxy and in thin-film transfer and heterogeneous integration, translating these capabilities into deployable, application-ready technologies demands close coordination with industrial partners. The UIUC team brings world-leading expertise in high-quality III–V epitaxial growth and in exfoliation and integration techniques that enable the creation of transferable device membranes without compromising material performance. However, the full value of these technologies can only be realized when coupled with application-driven design, system-level constraints, and manufacturing considerations led by industry. Korean SMEs possessing strong capabilities in device engineering, packaging, and product development, can directly leverage high-quality III–V epilayers and device membranes to create differentiated optoelectronic systems. A joint R&D framework enables rapid iteration between materials, devices, and applications, aligning fundamental advances with real-world performance targets. This collaboration is therefore essential to bridge the gap between epitaxial materials innovation and commercially relevant optoelectronic technologies.
Keywords	1. III–V epitaxy 2. Optoelectronic and photonic devices 3. III–V lift-off and integration 4. Transferrable device membranes

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5/6)
R&D Objectives	The proposed technology integrates complementary expertise in III–V epitaxial device growth and thin-film decoupling and integration. PI Lee contributes world-leading capabilities in III–V molecular beam epitaxy, enabling the growth of high-quality device epilayers with precise control over composition, interfaces, and optoelectronic functionality, suitable for emitters, detectors, and photovoltaic devices. In addition, he has a separate nitride MBE chamber equipped to grow novel materials including ferroelectric AlScN and superconductors like TaN, TiN, Ta, and Al with applications in memory and qubit technology. PI Kim provides advanced strategies for releasable growth templates that include sacrificial layers, chemical and mechanical releasing techniques, and integrating these epilayers for heterogeneous integration. Together, the team enables transferable III–V device membranes that preserve device performance while allowing heterogeneous integration onto diverse platforms. **1. Current research capability** **1.1. Molecular beam epitaxy of III-V devices** For this proposal, PI Lee will utilize his Veeco Mod Gen II III-V MBE system equipped for arsenide, phosphide, and antimonide growth. This chamber has grown a range of high-performance lasers including quantum well, quantum dot, and quantum cascade, covering wavelengths of 0.65-4.5 μm. The chamber has grown pioneering and world-record near-infrared $In_{0.53}Ga_{0.47}As$ detectors, including heterojunction phototransistors with ultra-high linear- and avalanche-mode gain at wavelengths up to 1.6 μm; these detectors can be modified to respond to wavelengths up to 2.2 μm. His group has extensive experience with device fabrication and multiple test setups for detectors and light emitters. **1.2. III-V layer release and integration** PI Kim aims to establish a robust and scalable pathway for realizing transferable III–V device			

epilayers and membranes that can be integrated onto diverse host platforms without compromising material or device quality. This effort leverages PI Kim's extensive expertise in 2D-materials-assisted epitaxy, epitaxial decoupling via mechanical lift-off and epitaxial lift-off, and heterogeneous integration, which collectively address long-standing limitations in conventional III–V device manufacturing. Central to this capability is the preparation of 2D material templates, including graphene and related atomically thin layers, that enable remote epitaxy with reduced defect formation, as well as mechanically robust exfoliation of epitaxial films. In parallel, sacrificial-layer-based approaches, such as AlAs-mediated epitaxial lift-off, provide deterministic and scalable routes for releasing device-quality membranes. By combining these approaches with integration strategies compatible with semiconductor manufacturing, this research aims to transform III–V epilayers from substrate-bound structures into transferable building blocks for advanced optoelectronic systems. The following sections describe the current capabilities underlying this effort and the proposed research objectives of the joint program.

Wafer-scale III-V membrane production by remote epitaxy. Within this framework, remote epitaxy exploits the ability of substrate lattice potentials to penetrate through a 2D material, such as graphene, allowing the overgrown film to inherit the crystallographic registry of the underlying substrate while remaining weakly bonded at the interface. This enables the growth of single-crystalline III–V epilayers that are structurally registered to the substrate yet can be mechanically exfoliated with minimal damage. It is important to note that exfoliation at the 2D interfaces can be achieved by simple mechanical lift-off process, while ensuring atomic precision of device layer peeling. These approaches therefore provide unprecedented control over interfacial energetics, defect evolution, and epilayer release, forming the foundation for transferable, high-quality III–V device membranes.

By leveraging these concepts, the PI has demonstrated wafer-scale remote epitaxy using not only crystalline graphene but also amorphous 2D materials, such as amorphous graphene (a-Gr) and amorphous boron nitride (a-BN), as practical and scalable growth templates. Unlike single-crystal graphene, which presents challenges for uniform growth on III–V substrates, amorphous 2D materials can be deposited conformally and uniformly on full wafers by MOCVD or MBE. The PI has shown that these amorphous 2D interlayers can still transmit sufficient substrate potential to enable remote epitaxial registry, while simultaneously providing chemically inert and mechanically weak interfaces. Using this approach, the PI has demonstrated remote epitaxy of both III–V and III–nitride semiconductors with high crystalline quality, followed by clean mechanical exfoliation of the epilayers and reuse of the underlying substrates. Importantly, the remote epitaxy process preserves substrate integrity, enabling multiple growth-release cycles and substantially reducing substrate cost. Beyond single-layer demonstrations, the PI has further realized multilayer 3D/2D stacked architectures, in which alternating stacks of epitaxial semiconductors and 2D release layers are grown and sequentially exfoliated. This multilayer strategy establishes a high-throughput pathway for producing multiple device-quality membranes from a single substrate by mechanical peeling, highlighting the scalability and manufacturing relevance of the approach for III–V and III–nitride device platforms.

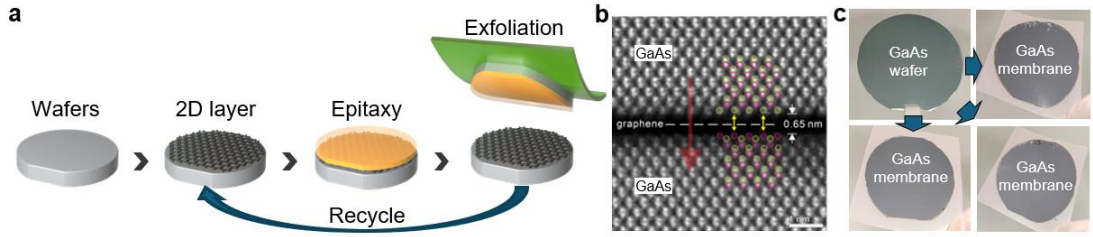

Figure 1. *Remote epitaxy for III-V membrane production.* ***a,*** *Schematic of remote epitaxy and III-V wafer recycling.* ***b,*** *GaAs grown on graphene/GaAs.* ***c,*** *Three GaAs membranes produced from a single GaAs wafer by remote epitaxy and substrate recycling.*

Wafer-scale III-V membrane production by selective-area epitaxy. The PI has also established selective-area epitaxy using 2D materials as growth masks, providing simultaneous control over nucleation, defect formation, and layer release. Because 2D materials are atomically thin and free of dangling bonds, they inherently suppress unwanted nucleation while allowing precise

definition of exposed growth windows through lithographic patterning or non-lithographic approaches such as ion bombardment. Epitaxy initiated within these nanoscale openings proceeds with strong crystallographic registration, while lateral overgrowth occurs across the surrounding 2D-covered regions. This geometry promotes effective strain accommodation at pattern edges and significantly suppresses the formation and propagation of threading dislocations compared with conventional selective-area or epitaxial lateral overgrowth techniques. Importantly, the presence of the 2D interlayer preserves a weakly bonded interface beneath the selectively grown regions, maintaining mechanical peelability of the epilayer after coalescence. As a result, large-area, single-crystalline films with reduced defect density can be obtained and subsequently exfoliated as intact membranes.

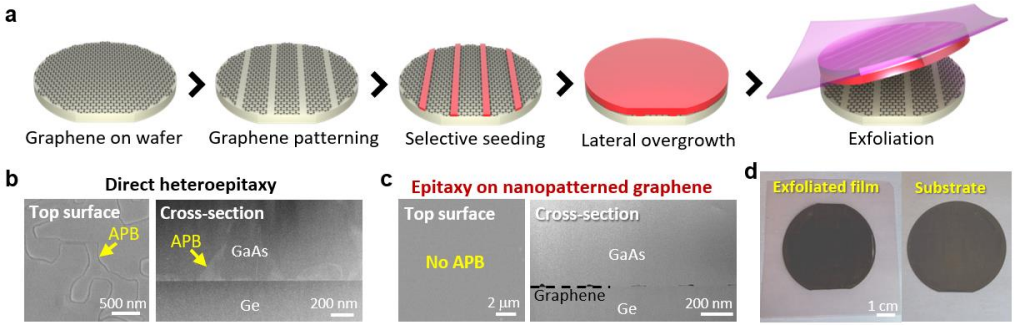

Figure 2. a, *Schematic of selective-area epitaxy utilizing 2D material mask.* ***b,c,*** *Top-view SEM (left) and cross-sectional TEM (right) showing complete elimination of APBs by involving nanopatterned graphene as an interlayer.* ***d,*** *Photograph showing 2-inch GaAs exfoliated from the substrate.*

III-V device platforms demonstrated by III-V membrane transfer. The PI has utilized the produced III-V membranes for functional platforms, using both 2D-assisted epitaxy and sacrificial layer-based epitaxial lift-off. One notable device platform enabled by these transfer technologies is vertically stacked light-emitting diodes (LEDs) formed from wafer-scale III–V LED membranes. In this approach, red, green, and blue LED epilayers are independently grown, released as intact membranes using 2D-assisted epitaxy or epitaxial lift-off, and sequentially transferred onto a single substrate. The stacked membranes are then vertically integrated and pixelated, allowing each color to be addressed within an extremely small footprint. The resulting vertically stacked RGB micro-LED platform achieves ultrahigh pixel densities that surpass previous limits set by lateral pixel layouts and conventional transfer techniques. This demonstration highlights the ability of membrane-based III–V device integration to overcome scaling constraints in high-resolution optoelectronic displays while maintaining uniformity and performance across large areas.

A second example is the development of flexible nitride membrane–based sensors for skin-mounted and wearable electronics. By exfoliating or lifting off thin III–nitride device layers and transferring them onto soft, deformable substrates, the PI has demonstrated mechanically compliant sensor platforms that maintain stable operation with multifunctionality under bending and stretching. These transferred nitride membranes enable intimate contact with textured surfaces (such as human skin) while retaining the intrinsic robustness of III–V performance, highlighting the versatility of device transfer approaches for both high-performance optoelectronics and flexible systems.

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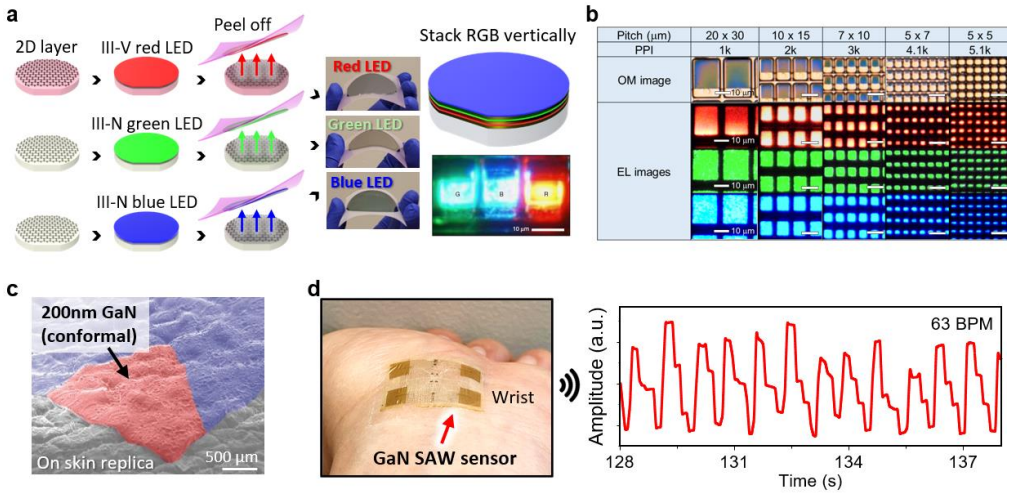

Figure 3. a, *Vertically stacked micro-LED by 2D materials-assisted epitaxy, exfoliation, and stacking.* ***b,*** *This approach enabled the world's smallest pixel size and the highest pixel density of 5100 ppi.* ***c,*** *SEM image of GaN exfoliated from 2D materials and integrated onto flexible platforms.* ***d,*** *Demonstration of pulse sensing by GaN SAW sensor made from remote epitaxial GaN.*

2. Proposed Research

2.1. III-V MBE growth on novel sacrificial and 2D release layers

PI Lee has extensive experience with growth of lattice-mismatched (i.e. metamorphic) devices and holds numerous records for his work on solar cells, detectors, and lasers, in many cases grown on silicon substrates. More recently, he has pioneered growth of defect-tolerant InP quantum dot light-emitting diodes and InAs quantum dot laser diodes on 2D release layers. In addition to extensive experience solving fundamental growth problems, his group fabricates benchmark devices on native substrates to compare performance of devices grown on unconventional substrates.

2.2. Heterogeneous integration of III-V devices

Throughout this project, we aim to enable fully controllable integration of III-V device membranes onto arbitrary platforms. First, we aim to improve the scalability and generality of heterogeneous integration by optimizing sacrificial-layer and 2D-material template preparation across a wide range of III–V substrates. While prior work has established the feasibility of using 2D materials as release layers and remote epitaxy templates, systematic control over their uniformity, thickness, defect density, and interfacial chemistry remains a key challenge for large-area manufacturing. We will develop substrate-agnostic processes for forming conformal 2D templates on representative III–V wafers, including arsenides and phosphides, and establish growth windows compatible with device-grade epitaxy. Emphasis will be placed on achieving wafer-scale uniformity, reproducibility across multiple growth-release cycles, and compatibility with existing epitaxial toolsets, enabling scalable production of transferable III–V device membranes.

Second, we will focus on improving control over the lift-off process itself, with particular attention to yield, reproducibility, and mechanical reliability. Current mechanical lift-off methods often rely on manual handling and empirically defined release conditions, which introduce variability and often lead to crack formation. We will investigate controlled crack initiation and propagation during membrane release, including the role of interfacial adhesion, membrane thickness, and stress distribution. Automation of the lift-off process will be explored through mechanical fixtures and process control strategies that enable deterministic, wafer-scale membrane separation. These efforts aim to suppress uncontrolled fracture, minimize edge damage, and establish repeatable lift-off protocols suitable for integration into manufacturing workflows. Similarly, we will investigate controlled epitaxial lift-off in chemicals by developing adequate support layers and handling processes.

Last, we aim to achieve the transfer and bonding of released III–V membranes onto target substrates. We will develop and compare multiple bonding and assembly techniques, including direct bonding, adhesive-assisted bonding, metal-assisted bonding, and hybrid approaches, to

	accommodate diverse host platforms and application requirements. Process variables such as surface preparation, bonding temperature, pressure, alignment accuracy, and interfacial cleanliness will be systematically studied to ensure mechanical integrity, thermal stability, and electrical continuity after transfer. By co-optimizing bonding strategies with membrane properties and target substrates, this research will establish reliable pathways for integrating III–V device membranes into complex, application-specific system architectures.
R&D Target Specifications	**1. Molecular beam epitaxy of III-V devices** • Growth on sacrificial layers and 2D template layers with RMS roughness below 2 nm across 50-100 mm wafers • Defect density within same order of magnitude as starting substrate • Wafer-scale thickness uniformity better than +/- 5% • Detectors: quantum efficiency within 10% of benchmark • Lasers: threshold current density within 2x of benchmark **2. III-V layer release and integration** • Wafer-scale sacrificial layer formation: Sacrificial layer's surface roughness < 1 nm at a wafer scale. • Wafer-scale 2D template formation: >99% continuous coverage without pinholes over full wafer area. • Lift-off yield: >95% intact III-V device membrane exfoliation at a wafer scale • Substrate recycling: Recycle the III-V substrate more than 3 times to demonstrate wafer reusability. • Transfer and bonding: Void-free and crack-free bonding of exfoliated III-V membranes onto target platforms.
R&D Plan	**Thrust 1. Molecular beam epitaxy of III-V devices** **Task 1.1 Growth on release layers and substrate reuse** In this task, we will develop high-quality and uniform III-V growth on sacrificial and 2D-template release layers developed by PI Kim. We will focus on initial nucleation conditions of both arsenide and phosphide templates that lead to smooth growth with minimal densities of extended defects. We will also extend these methods to growth on substrates where epitaxial layers have been exfoliated by PI Kim, paying attention to particle formation and accumulation of roughness and defects. **Milestones**: Growth on sacrificial layers and 2D template layers with RMS roughness below 2 nm across 50-100 mm wafers; defect density within same order of magnitude as starting substrate; wafer-scale thickness uniformity better than +/- 5% **Task 1.2 Device growth on release layers** In this task, we will develop growth of high-performance devices (e.g., detectors, lasers, transistors) on release layers and reused substrates and benchmark their performance with those on conventional substrates. **Milestones:** Detector quantum efficiency within 10%; for lasers (if desired), threshold current density within factor of 2 of benchmark. **Thrust 2. III-V layer release and integration** **Task 2.1: Scalable sacrificial-layer and 2D-template preparation** We develop wafer-scale sacrificial layers and 2D-material templates with controlled roughness, continuity, and interfacial chemistry across multiple III–V substrates. Process development focuses on arsenide- and phosphide-based wafers using deposition conditions compatible with device-grade epitaxy. Surface roughness, thickness uniformity, and pinhole density are characterized at the wafer scale using AFM, optical inspection, and cross-sectional analysis. Process repeatability is evaluated through multiple growth-release cycles on the same substrate to assess long-term stability and

	compatibility with epitaxial toolsets. **Milestones:** Sacrificial-layer surface roughness below 1 nm RMS across full wafers; greater than 99% continuous 2D-template coverage without pinholes; repeatable template formation over multiple growth-release cycles. **Task 2.2: Controlled III-V device layer lift-off processes** We improve lift-off yield and reproducibility by controlling crack initiation, crack propagation, and membrane handling during release. Mechanical lift-off studies will be systematically conducted to quantify the influence of membrane thickness, residual stress, and interfacial adhesion on fracture behavior. Controlled crack paths are introduced to suppress uncontrolled edge fracture and improve intact membrane yield. Epitaxial lift-off is investigated in parallel through optimized sacrificial-layer designs, temporary support layers, and wet-handling protocols that preserve membrane integrity. Handling strategies are evaluated using guided fixtures and process control schemes to reduce operator-dependent variability. **Milestones:** Greater than 95% intact III–V membrane lift-off yield at wafer scale; suppression of uncontrolled cracking during release; successful recycling of III–V substrates more than three times. **Task 2.3: Transfer and bonding onto target platforms** We will establish reliable transfer and bonding routes for released III–V membranes onto diverse target substrates. Direct bonding, adhesive-assisted bonding, metal-assisted bonding, and hybrid approaches are systematically compared. Surface preparation, bonding temperature, applied pressure, alignment accuracy, and interfacial cleanliness are controlled to achieve mechanically stable and electrically viable interfaces. Bonded membranes are inspected for voids and cracks and evaluated for mechanical integrity after transfer. **Milestones:** Void-free and crack-free bonding of III–V membranes onto target platforms; mechanically stable adhesion suitable for downstream device processing and system-level integration.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The PIs at UIUC will be responsible for the development and validation of high-quality III–V epitaxial materials and transferable device membranes. We will design and grow device-grade III–V epilayers using molecular beam epitaxy, with controlled composition, doping, and interfaces tailored for optoelectronic functionality. We will develop and implement sacrificial-layer-based exfoliation, such as epitaxial lift-off or 2D-material-assisted approaches, to produce freestanding or transferable III–V device membranes while preserving crystalline and device integrity. We will perform on-wafer material and device characterization to evaluate structural, optical, and electrical performance. We will also be responsible for ensuring membrane production without performance degradation. The device samples will be shared with Korean SME partners for subsequent development of integrated platforms.
Expected R&R of Korean SME Partner	Korean SME partners will be responsible for application-driven development and system-level utilization of the III–V device epilayers and transferable device membranes provided through the joint R&D project. SMEs will define target applications and performance requirements based on their product roadmaps, such as sensing, imaging, energy harvesting, or integrated optoelectronic systems. They will lead device adaptation, packaging, module integration, and reliability evaluation under application-relevant conditions. Korean SMEs will also contribute to process optimization from a manufacturing and scalability perspective, providing feedback on integration compatibility, yield, and cost considerations. During the joint project, SMEs will collaborate closely with the research institute to refine device specifications, co-develop prototypes, and demonstrate application-specific functionality. SMEs will further support system-level validation and prepare pathways toward commercialization, ensuring that the jointly developed technologies are aligned

	with real-world deployment needs.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint collaboration is expected to directly strengthen the technological capabilities and commercialization potential of participating Korean SMEs. By accessing high-quality III–V device epilayers and transferable device membranes, SMEs will be able to develop advanced optoelectronic components that are difficult to realize using existing manufacturing approaches. The project will enable SMEs to rapidly prototype and validate application-specific devices and modules, accelerating the transition from R&D to product development. Technological outcomes include improved device performance, expanded integration options, and reduced reliance on expensive III–V substrates, leading to lower production costs and greater design flexibility. These advances are expected to support the creation of differentiated products in areas such as sensing, imaging, LiDAR, energy harvesting, and emerging intelligent systems. From a broader perspective, the collaboration will enhance the global competitiveness of Korean SMEs, promote the development of high-value optoelectronic technologies within Korea, and contribute to long-term industrial growth by establishing pathways for scalable manufacturing and international technology partnerships.